



Digital Integrated Circuit Design

Lecture-7: CORDIC

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Outline

- ☐ Rotations and Pseudorotations
- ☐ Basic CORDIC Iterations
- ☐ CORDIC Hardware
- ☐ Generalized CORDIC

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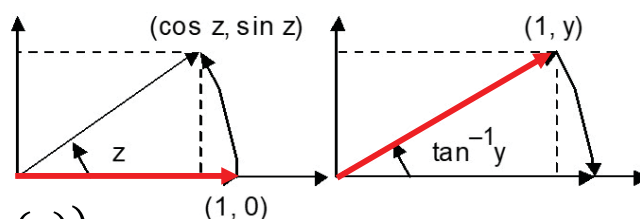
COordinate Rotation DIgital Computer (CORDIC)

❑ CORIDC

- Evaluation of **trigonometric**, **hyperbolic**, and other common functions, such as **log**, **exp**, etc.
- Low-cost method that achieves the reasonable accuracy of about 1-bit, but does not guarantee proper rounding

❑ Key ideas for CORDIC

- Point-A = $(x, y) = (1, 0)$
- Rotate point-A with $\theta = z$
- End-point of A at $(\cos(z), \sin(z))$



❑ A computationally efficient way of rotating a vector

- Easily evaluate $\cos$, $\sin$, and $\tan^{-1}$ functions

Rotations and Pseudorotations

❑ Difficult to rotate an arbitrary angle

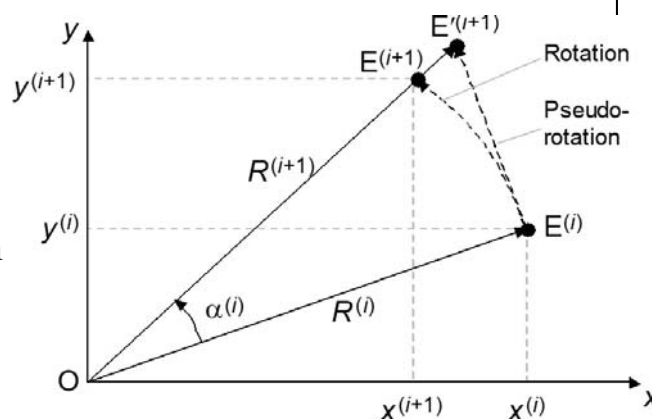
❑ Perform pseudorotations with simpler operations

- Use special angles to synthesize the desired angle z

$$z = \alpha^{(1)} + \alpha^{(2)} + \dots + \alpha^{(m)}$$

❑ Real- and pseudo-rotation

- Start-point: $E^{(i)}$ at coordinates $(x^{(i)}, y^{(i)})$
- End-point: $E'^{(i+1)}$ at coordinates $(x'^{(i+1)}, y'^{(i+1)})$ in pseudorotation
- End-point: $E^{(i+1)}$ at coordinates $(x^{(i+1)}, y^{(i+1)})$ in real rotation



Rotations and Pseudorotations (cont'd)

Real rotation

$$x^{(i+1)} = x^{(i)} \cos \alpha^{(i)} - y^{(i)} \sin \alpha^{(i)} = \frac{x^{(i)} - y^{(i)} \tan \alpha^{(i)}}{(1 + \tan^2 \alpha^{(i)})^{1/2}}$$

$$y^{(i+1)} = y^{(i)} \cos \alpha^{(i)} + x^{(i)} \sin \alpha^{(i)} = \frac{y^{(i)} + x^{(i)} \tan \alpha^{(i)}}{(1 + \tan^2 \alpha^{(i)})^{1/2}}$$

$$z^{(i+1)} = z^{(i)} - \alpha_i$$

■ $z^{(i)}$: the residual rotation

Pseudorotation

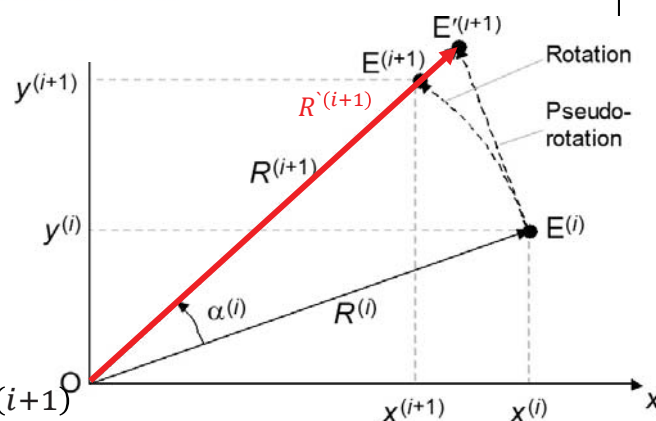
$$x^{(i+1)} = x^{(i)} - y^{(i)} \tan \alpha^{(i)}$$

$$y^{(i+1)} = y^{(i)} + x^{(i)} \tan \alpha^{(i)}$$

$$z^{(i+1)} = z^{(i)} - \alpha^{(i)}$$

■ Increase length from $R^{(i)}$ to $R^{(i+1)}$

■ Changed the length $R^{(i+1)}$ in pseudorotation



5

Rotations and Pseudorotations (cont'd)

Pseudorotation

$$R^{(i+1)} = R^{(i)} / \cos \alpha^{(i)} = R^{(i)} (1 + \tan^2 \alpha^{(i)})^{1/2}$$

Real rotation

■ Assume m real rotations by $\alpha^{(1)}, \alpha^{(2)}, \dots, \alpha^{(m)}$, given $x^{(0)} = x, y^{(0)} = y$, and $z^{(0)} = z$

■ After m real rotations, we have

$$x^{(m)} = x \cos(\sum \alpha^{(i)}) - y \sin(\sum \alpha^{(i)})$$

$$y^{(m)} = y \cos(\sum \alpha^{(i)}) + x \sin(\sum \alpha^{(i)})$$

$$z^{(m)} = z - (\sum \alpha^{(i)})$$

Rotations and Pseudorotations (cont'd)

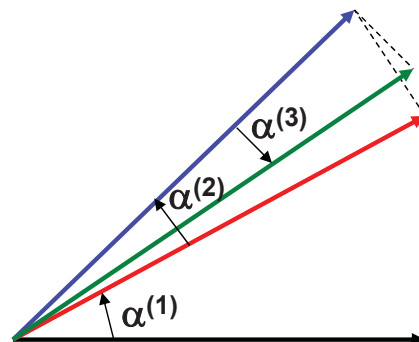
❑ Pseudorotation step in CORDIC

- Assume m real rotations by $\alpha^{(1)}, \alpha^{(2)}, \dots, \alpha^{(m)}$, given $x^{(0)} = x, y^{(0)} = y$, and $z^{(0)} = z$
- After m pseudorotations, we have

$$x^{(m)} = K[x \cos(\sum \alpha^{(i)}) - y \sin(\sum \alpha^{(i)})]$$

$$y^{(m)} = K[y \cos(\sum \alpha^{(i)}) + x \sin(\sum \alpha^{(i)})]$$

$$z^{(m)} = z - (\sum \alpha^{(i)})$$
- $K = \prod (1 + \tan^2 \alpha^{(i)})^{1/2}$
- K is a **constant** if angles of rotation are always the same, differing only in **sign** or **direction**.



Basic CORDIC Iterations

❑ Basic CORDIC iteration

- In step i , the pseudorotation performs by an angle $\tan(e^i) = 2^{-i}$, and $d_i = \text{sign}(z^i)$ where $d_i \in [-1, 1]$.

$x^{(i+1)} = x^{(i)} - d_i y^{(i)} 2^{-i}$	$x^{(i+1)} = x^{(i)} - y^{(i)} \tan \alpha^{(i)}$
$y^{(i+1)} = y^{(i)} + d_i x^{(i)} 2^{-i}$	$y^{(i+1)} = y^{(i)} + x^{(i)} \tan \alpha^{(i)}$
$z^{(i+1)} = z^{(i)} - d_i \tan^{-1} 2^{-i}$	$z^{(i+1)} = z^{(i)} - \alpha^{(i)}$

i	$e^{(i)} \text{deg.}$	$e^{(i)} \text{rad.}$
0	45.0	0.785 398 163
1	26.6	0.463 647 609
2	14.0	0.244 978 663
3	7.1	0.124 354 994
4	3.6	0.062 418 810
5	1.8	0.031 239 833
6	0.9	0.015 623 728
7	0.4	0.007 812 341
8	0.2	0.003 906 230
9	0.1	0.001 953 123

- $e^{(i)}$: precomputed and stored in a table
- $\alpha^{(i)} = d_i \tan^{-1}(2^{-i})$

■ Make z converge to 0

❑ For example: 30° angle

$$\begin{aligned}
 30.0 &\cong 45.0 - 26.6 + 14.0 - 7.1 + 3.6 + 1.8 \\
 &\quad - 0.9 + 0.4 - 0.2 + 0.1 \\
 &= 30.1
 \end{aligned}$$

$$\begin{aligned}
 x^{(i+1)} &= x^{(i)} - y^{(i)} \tan \alpha^{(i)} \\
 y^{(i+1)} &= y^{(i)} + x^{(i)} \tan \alpha^{(i)} \\
 z^{(i+1)} &= z^{(i)} - \alpha^{(i)}
 \end{aligned}$$

Basic CORDIC Iterations (cont'd)

- Choosing the angles to force z to zero by choosing $d_i = \text{sign}(z^{(i)})$

$$x^{(i+1)} = x^{(i)} - d_i y^{(i)} 2^{-i}$$

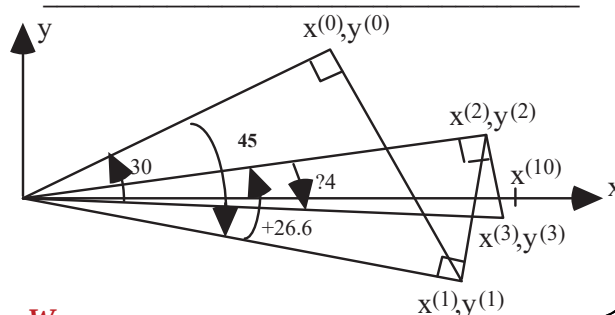
$$y^{(i+1)} = y^{(i)} + d_i x^{(i)} 2^{-i}$$

$$z^{(i+1)} = z^{(i)} - d_i \tan^{-1} 2^{-i}$$

$$= z^{(i)} - d_i e^{(i)}$$

- For k bits of precision in results, k CORDIC iterations are needed, because $\tan^{-1} 2^{-i} \cong 2^{-i}$ for large i .

i	$z^{(i)}$	$-d_i e^{(i)}$	$= z^{(i+1)}$
0	+30.0	- 45.0	+30.0
1	-15.0	+ 26.6	+11.6
2	+11.6	- 14.0	-2.4
3	-2.4	+ 7.1	+4.7
4	+4.7	- 3.6	+1.1
5	+1.1	- 1.8	-0.7
6	-0.7	+ 0.9	+0.2
7	+0.2	- 0.4	-0.2
8	-0.2	+ 0.2	+0.0
9	+0.0	- 0.1	-0.1



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9

Basic CORDIC Iterations (cont'd)

- Convergent problem

■ $-99.7^\circ \leq z \leq 99.7^\circ$

■ $\pm \sum e^i (\text{deg}) \Rightarrow \approx \pm 99.7^\circ$

■ $\pm \sum e^i \rightarrow \pm 1.7433 (\text{rad}), \text{ when } i \rightarrow \infty$

- In practical application

■ $-90^\circ \leq z \leq 90^\circ$, i.e., $[-\pi/2, \pi/2]$ radians

- Quadrant mapping for out of $[-\pi/2, \pi/2]$

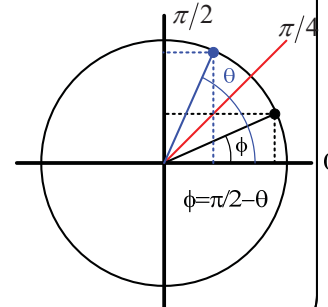
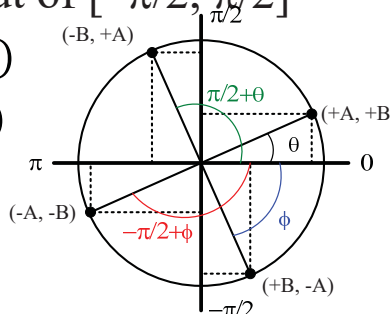
■ $\cos(z \pm 2n\pi) = \cos(z)$

■ $\sin(z \pm 2n\pi) = \sin(z)$

■ $\cos(z - \pi) = -\cos(z)$

■ $\sin(z - \pi) = -\sin(z)$

i	$e^{(i)} \text{deg.}$	$e^{(i)} \text{rad.}$
0	45.0	0.785 398 163
1	26.6	0.463 647 609
2	14.0	0.244 978 663
3	7.1	0.124 354 994
4	3.6	0.062 418 810
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7	0.4	0.007 812 341
8	0.2	0.003 906 230
9	0.1	0.001 953 123



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10

Basic CORDIC Iterations (cont'd)

□ Consideration of K

■ $K = \prod (1 + \tan^2 \alpha^{(i)})^{1/2}$

■ $K = 1.646760258121$

$$K = \prod \sqrt{1 + \tan^2(\alpha^i)}$$

$$= \prod \sqrt{1 + 2^{-2i}} = 1.6467$$

$1/K \approx 0.60725$

$$= 2^{-1} \left[\begin{array}{l} (1 + 2^{-2}) \cdot (1 - 2^{-5}) \cdot \\ (1 + 2^{-8}) \cdot (1 + 2^{-10}) \end{array} \right]$$

Iteration	K	$1/K$
0	1.4142135623730951	0.7071067811865475
1	1.5811388300841898	0.6324555320336759
2	1.6298006013006623	0.6135719910778963
3	1.6424840657522373	0.6088339125177524
4	1.6456889157572550	0.6076482562561681
5	1.6464922787124787	0.6073517701412959
6	1.6466932542736437	0.6072776440935260
7	1.6467435065969009	0.6072591122988928
8	1.6467560702048785	0.6072544793325624
9	1.6467592111398222	0.6072533210898752
10	1.6467599963756174	0.6072530315291344
11	1.6467601926846949	0.6072529591389448
12	1.6467602417619722	0.6072529410413973
13	1.6467602540312922	0.6072529365170103
14	1.6467602570986222	0.6072529353859135
15	1.6467602578654548	0.6072529351031394
16	1.6467602580571630	0.6072529350324457
17	1.6467602581050900	0.6072529350147724
18	1.6467602581170717	0.6072529350103540
19	1.6467602581200671	0.6072529350092495

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11

Rotation Mode

□ Evaluation using K to find $\cos(z)$ and $\sin(z)$

■ Let $x = 1/K = 0.607\ 252\ 935 \dots$ and $y = 0$

$$x^{(i+1)} = x^{(i)} - d_i y^{(i)} 2^{-i}$$

$$y^{(i+1)} = y^{(i)} + d_i x^{(i)} 2^{-i}$$

$$z^{(i+1)} = z^{(i)} - d_i \tan^{-1} 2^{-i}$$

$$= z^{(i)} - d_i e^{(i)}$$

$$x^{(m)} = \cancel{K}(\cancel{x} \cos z - y \overset{0}{\sin z}) = \cos z$$

$$y^{(m)} = \cancel{K}(\overset{0}{y} \cos z + \cancel{x} \sin z) = \sin z$$

$$z^{(m)} = \overset{0}{0}$$

$$K = 1.646\ 760\ 258\ 121 \dots$$

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12

Vectoring Mode

□ Make y tend to zero by choosing $d_i = -\text{sign}(x^{(i)}y^{(i)})$ or $d_i = -\text{sign}(y^{(i)})$

□ After m iterations, we have $\tan(\sum \alpha^i) = y/x$

$$x^{(m)} = K \left[x \cos \left(\sum \alpha^{(i)} \right) - y \sin \left(\sum \alpha^{(i)} \right) \right]$$

$$= \frac{K (x - y \tan (\sum \alpha^{(i)}))}{[1 + \tan^2 (\sum \alpha^{(i)})]^{1/2}}$$

$$= \frac{K (x + y^2/x)}{(1 + y^2/x^2)^{1/2}}$$

$$= K(x^2 + y^2)^{1/2}$$

□ Vectoring mode in CORDIC equation

$$x^{(m)} = K(x^2 + y^2)^{1/2}$$

$$y^{(m)} = 0$$

$$z^{(m)} = z + \tan^{-1}(y/x)$$

$$K = 1.646\ 760\ 258\ 121\ \dots$$

$$x^{(i+1)} = x^{(i)} - d_i y^{(i)} 2^{-i}$$

$$y^{(i+1)} = y^{(i)} + d_i x^{(i)} 2^{-i}$$

$$z^{(i+1)} = z^{(i)} - d_i \tan^{-1} 2^{-i}$$

$$x^{(i+1)} = x^{(i)} - y^{(i)} \tan \alpha^{(i)}$$

$$y^{(i+1)} = y^{(i)} + x^{(i)} \tan \alpha^{(i)}$$

$$z^{(i+1)} = z^{(i)} - \alpha^{(i)}$$

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13

Vectoring Mode (cont'd)

□ Evaluation to find $\tan^{-1}y$

■ Let $x = 1$ and $z = 0$

$$x^{(m)} = K(x^2 + y^2)^{1/2}$$

$$y^{(m)} = 0$$

$$z^{(m)} = z + \tan^{-1}(y/x)$$

$$K = 1.646\ 760\ 258\ 121\ \dots$$

□ $\tan^{-1}(1/y) = \pi/2 - \tan^{-1}y$

□ Other triangular functions

■ $\tan z$ obtained from $\sin z$ and $\cos z$ via division

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14

CORDIC Hardware

CORDIC equations

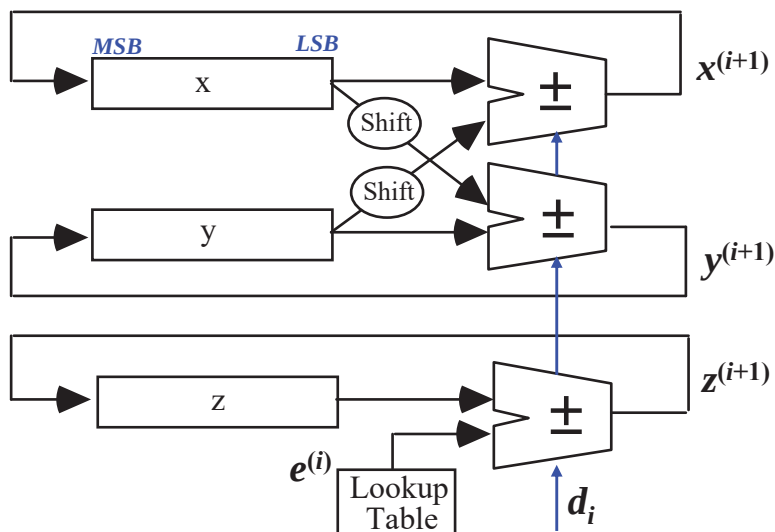
$$x^{(i+1)} = x^{(i)} - d_i y^{(i)} 2^{-i}$$

$$y^{(i+1)} = y^{(i)} + d_i x^{(i)} 2^{-i}$$

$$z^{(i+1)} = z^{(i)} - d_i \tan^{-1} 2^{-i}$$

$$= z^{(i)} - d_i e^{(i)}$$

- k table entries for k bits of precision



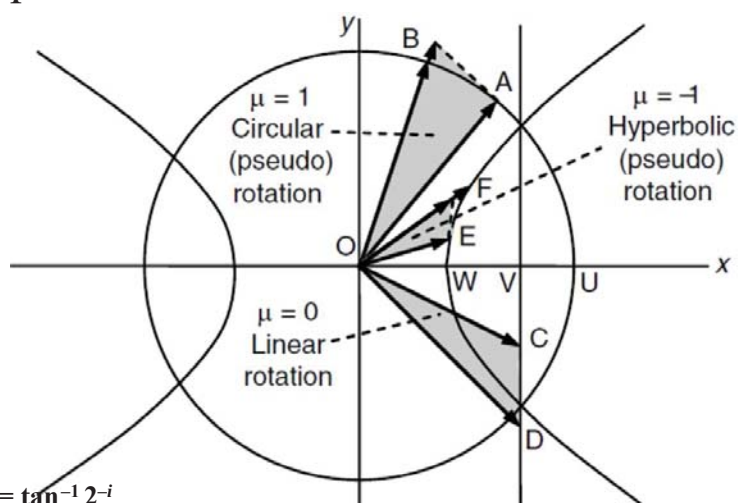
Generalized CORDIC

Generalized CORDIC equations

$$x^{(i+1)} = x^{(i)} - \mu d_i y^{(i)} 2^{-i}$$

$$y^{(i+1)} = y^{(i)} + d_i x^{(i)} 2^{-i}$$

$$z^{(i+1)} = z^{(i)} - d_i e^{(i)}$$



$\mu = 1$ Circular rotations
(basic CORDIC)

$\mu = 0$ Linear rotations

$\mu = -1$ Hyperbolic rotations

$$e^{(i)} = \tan^{-1} 2^{-i}$$

$$e^{(i)} = 2^{-i}$$

$$e^{(i)} = \tanh^{-1} 2^{-i}$$

Summary Generalized CORDIC

Mode →	Rotation: $d_i = \text{sign}(z^{(i)})$, $z^{(i)} \rightarrow 0$	Vectoring: $d_i = -\text{sign}(x^{(i)}y^{(i)})$, $y^{(i)} \rightarrow 0$
$\mu = 1$ Circular $e^{(i)} = \tan^{-1} 2^{-i}$	$\begin{array}{c} x \rightarrow \text{CORDIC} \rightarrow K(x \cos z - y \sin z) \\ y \rightarrow \text{CORDIC} \rightarrow K(y \cos z + x \sin z) \\ z \rightarrow \text{CORDIC} \rightarrow 0 \end{array}$ For cos & sin, set $x = 1/K$, $y = 0$ $\tan z = \sin z / \cos z$	$\begin{array}{c} x \rightarrow \text{CORDIC} \rightarrow K \sqrt{x^2 + y^2} \\ y \rightarrow \text{CORDIC} \rightarrow 0 \\ z \rightarrow \text{CORDIC} \rightarrow z + \tan^{-1}(y/x) \end{array}$ For $\tan^{-1}$, set $x = 1$, $z = 0$ $\cos^{-1} w = \tan^{-1}[\sqrt{1-w^2}/w]$ $\sin^{-1} w = \tan^{-1}[w/\sqrt{1-w^2}]$
$\mu = 0$ Linear $e^{(i)} = 2^{-i}$	$\begin{array}{c} x \rightarrow \text{CORDIC} \rightarrow x \\ y \rightarrow \text{CORDIC} \rightarrow y + xz \\ z \rightarrow \text{CORDIC} \rightarrow 0 \end{array}$ For multiplication, set $y = 0$	$\begin{array}{c} x \rightarrow \text{CORDIC} \rightarrow x \\ y \rightarrow \text{CORDIC} \rightarrow 0 \\ z \rightarrow \text{CORDIC} \rightarrow z + y/x \end{array}$ For division, set $z = 0$
$\mu = -1$ Hyperbolic $e^{(i)} = \tanh^{-1} 2^{-i}$	$\begin{array}{c} x \rightarrow \text{CORDIC} \rightarrow K'(x \cosh z - y \sinh z) \\ y \rightarrow \text{CORDIC} \rightarrow K'(y \cosh z + x \sinh z) \\ z \rightarrow \text{CORDIC} \rightarrow 0 \end{array}$ For cosh & sinh, set $x = 1/K'$, $y = 0$ $\tanh z = \sinh z / \cosh z$ $\exp(z) = \sinh z + \cosh z$ $w^t = \exp(t \ln w)$	$\begin{array}{c} x \rightarrow \text{CORDIC} \rightarrow K' \sqrt{x^2 - y^2} \\ y \rightarrow \text{CORDIC} \rightarrow 0 \\ z \rightarrow \text{CORDIC} \rightarrow z + \tanh^{-1}(y/x) \end{array}$ For $\tanh^{-1}$, set $x = 1$, $z = 0$ $\ln w = 2 \tanh^{-1} (w-1)/(w+1)$ $\sqrt{w} = \sqrt{(w+1/4)^2 - (w-1/4)^2}$ $\cosh^{-1} w = \ln(w + \sqrt{1-w^2})$ $\sinh^{-1} w = \ln(w + \sqrt{1+w^2})$
Note →	In executing the iterations for $\mu = -1$, steps 4, 13, 40, 121, $\dots$, j , $3j+1$, $\dots$ must be repeated. These repetitions are incorporated in the constant K' below.	

$$\mu \in \{-1, 0, 1\}$$

$$d_i \in \{-1, 1\}$$

$$K = 1.646\ 760\ 258\ 121\ \dots$$

$$1/K = 0.607\ 252\ 935\ 009\ \dots$$

$$K' = 0.828\ 159\ 360\ 960\ 2\ \dots$$

$$1/K' = 1.207\ 497\ 067\ 763\ \dots$$

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17

CORDIC Design

Equations

$$x_{i+1} = x_i - \mu d_i y_i 2^{-i}$$

$$y_{i+1} = y_i + d_i x_i 2^{-i}$$

$$z_{i+1} = z_i - d_i e_i$$

$$\mu = 1, \quad \text{Circular rotations} \quad e_i = \tan^{-1} 2^{-i}$$

$$\mu = 0, \quad \text{Linear rotations} \quad e_i = 2^{-i}$$

$$\text{Rotation mode} \Rightarrow d_i = \text{sign}(z_i), \quad z_i \rightarrow 0$$

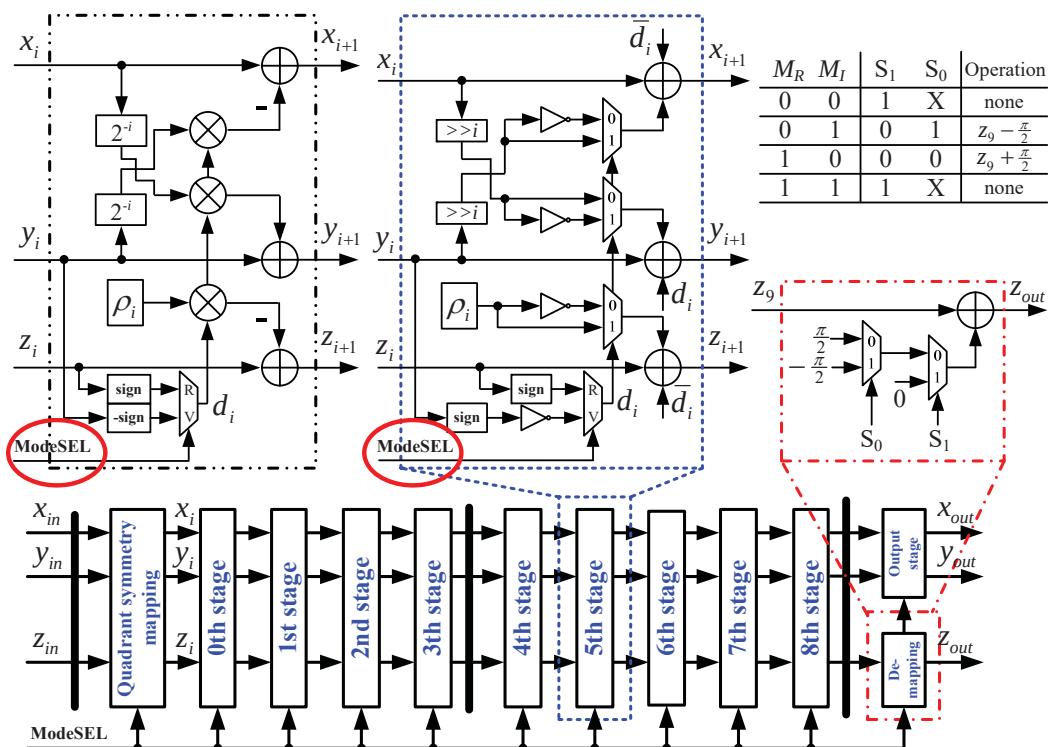
$$\text{Vectoring mode} \Rightarrow d_i = -\text{sign}(x_i \cdot y_i), \quad y_i \rightarrow 0$$

$$\Rightarrow d_i = -\text{sign}(y_i), \quad y_i \rightarrow 0$$

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18

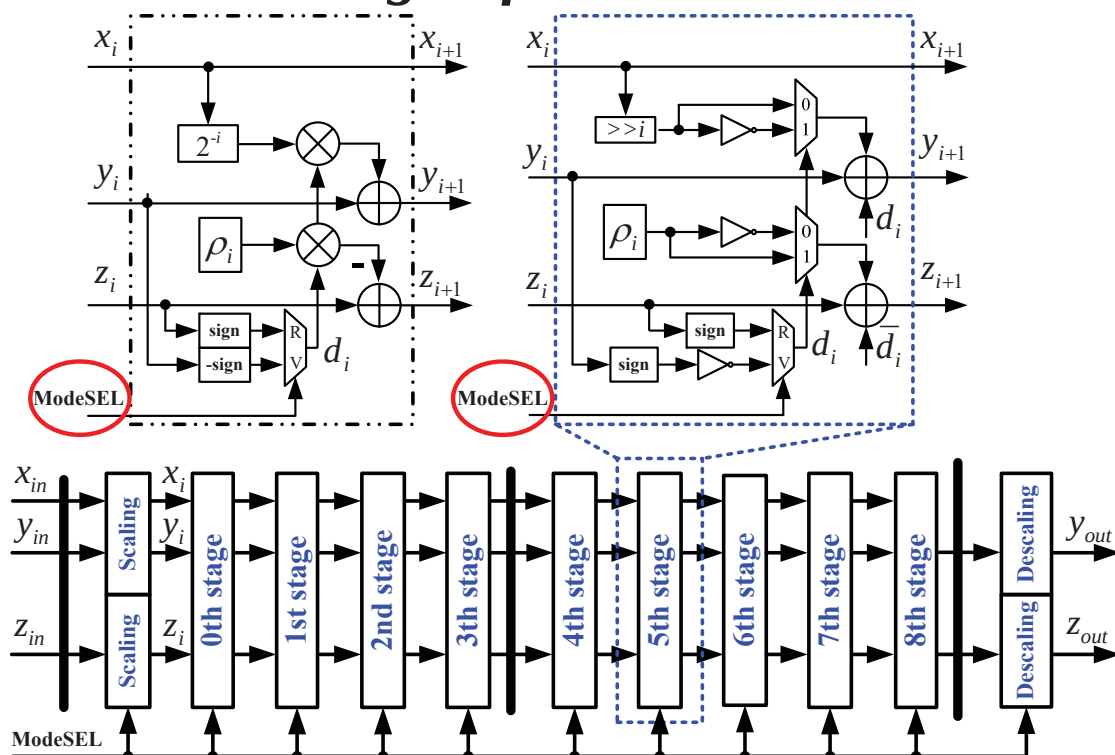
Hardware Design of CRM/CVM Module



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19

Hardware Design of LRM/LVM Module



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20

Linear CORDIC–Rotation Mode

- A linear rotation by angle $\theta_i = 2^{-i}$

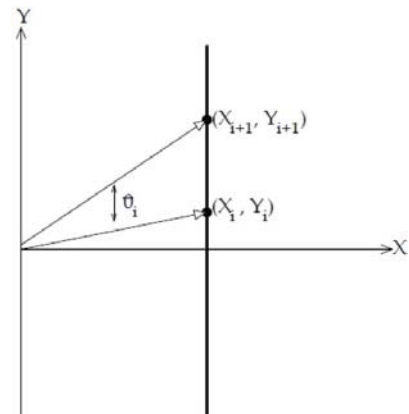
$$x_{i+1} = x_i$$

$$y_{i+1} = y_i + d_i x_i 2^{-i}$$

$$z_{i+1} = z_i - d_i \theta_i$$

$$d_i = \text{sign}(z_i)$$

$$\theta_i = 2^{-i} \quad i = 0, 1, 2, 3, \dots$$



- For example, $x_0 = x$, $y_0 = y$, $z_0 = z$, after $n+1$ iterations, we have

- $y_{n+1} \approx y_0 + x_0 z_0$

- $z_{n+1} \approx 0$

- For the special case, $y_0 = 0$, then $y_{n+1} \approx x_0 z_0$.

Linear CORDIC–Rotation Mode (cont'd)

- Multiply-Add

- $y_{n+1} = y_0 + \sum_{i=0}^n d_i 2^{-i} x_i$

$$= y_0 + x_0 \sum_{i=0}^n d_i 2^{-i}$$

- $z_{n+1} = z_0 - \sum_{i=0}^n d_i 2^{-i} x_i$

- Let $z_0 \approx \sum_{i=0}^n d_i 2^{-i}$ and $z_{n+1} \approx 0$

- Final, $y_{n+1} \approx y_0 + x_0 z_0$

- The range of z_0 : $|z_0| \leq 2$

Linear CORDIC–Rotation Mode (cont'd)

□ $x = 0.37, y = 0.48, xz = 0.1776$

i	X_i	Y_i	Z_i	d_i
0	0.0101111010111000	0.0000000000000000	0.0111101011100001	1
1	0.0101111010111000	0.0101111010111000	-0.1000010100011110	-1
2	0.0101111010111000	0.0010111101011100	-0.0000010100011110	-1
3	0.0101111010111000	0.0001011110101110	0.0011101011100001	1
4	0.0101111010111000	0.0010001110000101	0.0001101011100001	1
5	0.0101111010111000	0.0010100101110000	0.0000101011100001	1
6	0.0101111010111000	0.0010110001100110	0.0000001011100001	1
7	0.0101111010111000	0.0010110111100001	-0.0000000100011110	-1
8	0.0101111010111000	0.0010110100100011	0.0000000011100001	1
9	0.0101111010111000	0.0010110110000010	-0.0000000000011110	-1
10	0.0101111010111000	0.0010110101010011	0.0000000001100001	1
11	0.0101111010111000	0.0010110101101010	0.0000000000100001	1
12	0.0101111010111000	0.0010110101110110	0.0000000000000001	1
13	0.0101111010111000	0.0010110101111100	-0.0000000000001110	-1
14	0.0101111010111000	0.0010110101111001	-0.0000000000000110	-1
15	0.0101111010111000	0.0010110101111000	-0.0000000000000010	-1

$x \times z \approx 0.1776$

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23

Linear CORDIC–Vectoring Mode

□ A linear rotation by angle $\theta_i = 2^{-i}$

$$x_{i+1} = x_i$$

$$y_{i+1} = y_i + d_i x_i 2^{-i}$$

$$z_{i+1} = z_i - d_i \theta_i$$

$$d_i = -\text{sign}(y_i)$$

$$\theta_i = 2^{-i} \quad i = 0, 1, 2, 3, \dots$$

□ For example, $x_0 = x, y_0 = y, z_0 = 0$, after $n+1$ iterations, we have

■ $z_{n+1} \approx z_0 + y_0/x_0$

■ $y_{n+1} \approx 0$

■ For the special case, $z_0 = 0$, then $y_{n+1} \approx y_0/x_0$.

■ The range: $|y_0/x_0| \leq 2$

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24

Linear CORDIC–Vectoring Mode

□ For the special case $z_0 = 0$, we have

■ $y_0 + x_0 \sum_{i=0}^n d_i 2^{-i} x_i \approx 0$

$$\sum_{i=0}^n d_i 2^{-i} x_i = -\frac{y_0}{x_0}$$

■ $z_{n+1} = z_0 - \sum_{i=0}^n d_i 2^{-i} x_i$

$$z_{n+1} = z_0 + \frac{y_0}{x_0}$$

Hyperbolic

□ Definition for Hyperbolic

■ $\sinh x = \frac{e^x - e^{-x}}{2}$

■ $\cosh x = \frac{e^x + e^{-x}}{2}$

■ $\tanh x = \frac{\sinh x}{\cosh x}$

■ $\coth x = \frac{1}{\tanh x}$

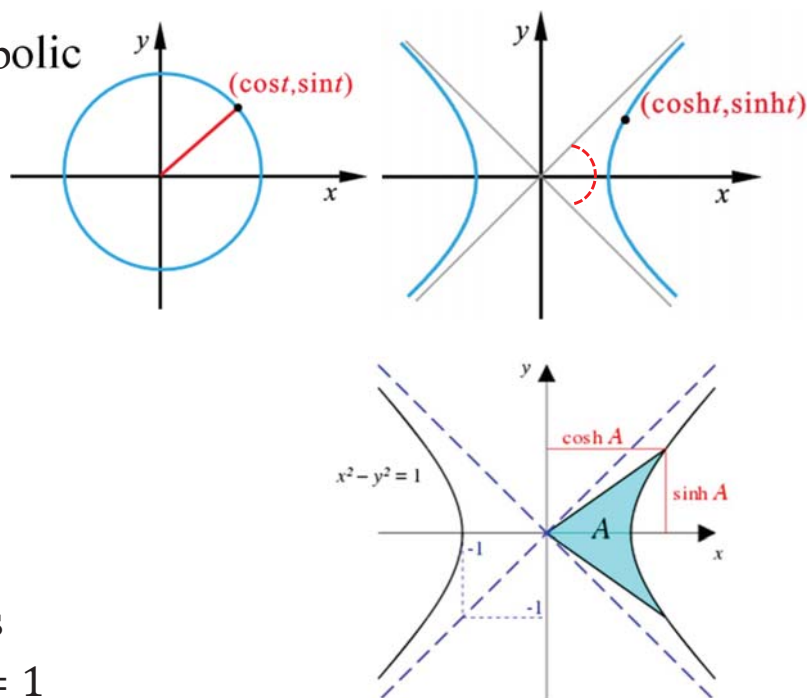
■ $\operatorname{sech} x = \frac{1}{\cosh x}$

■ $\operatorname{csch} x = \frac{1}{\sinh x}$

□ The related equations

■ $\cosh^2 x - \sinh^2 x = 1$

■ $\operatorname{sech}^2 x = 1 - \tanh^2 x$



Hyperbolic CORDIC – Rotation Mode

□ An Hyperbolic rotation by angle θ_i

$$\begin{bmatrix} x_{i+1}^* \\ y_{i+1}^* \end{bmatrix} = \begin{bmatrix} \cosh\theta_i & \sinh\theta_i \\ \sinh\theta_i & \cosh\theta_i \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix}$$

$$\blacksquare x_{i+1}^* = x_i \cosh\theta_i + y_i \sinh\theta_i$$

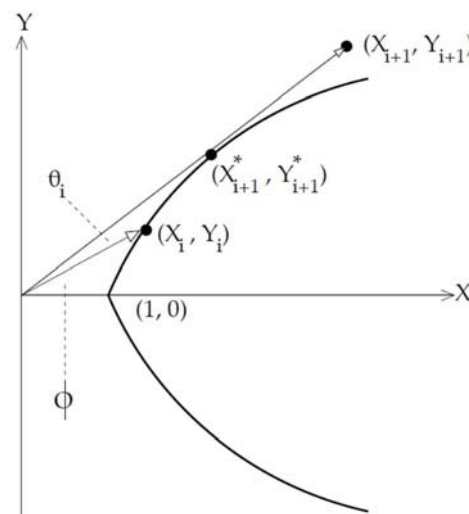
$$\blacksquare y_{i+1}^* = y_i \cosh\theta_i + x_i \sinh\theta_i$$

$$\blacksquare x_{i+1}^* = \cosh\theta_i \left[x_i + y_i \frac{\sinh\theta_i}{\cosh\theta_i} \right]$$

$$\blacksquare y_{i+1}^* = \cosh\theta_i \left[y_i + x_i \frac{\sinh\theta_i}{\cosh\theta_i} \right]$$

$$\blacksquare \operatorname{sech}^2\theta_i = 1 - \tanh^2\theta_i$$

$$\blacksquare \cosh\theta_i = \frac{1}{\operatorname{sech}\theta_i}$$



Hyperbolic CORDIC – Rotation Mode (cont'd)

□ An Hyperbolic rotation by angle θ

$$\blacksquare x_{i+1} = x_{i+1}^* / \cosh\theta_i = x_i + y_i \tanh\theta_i$$

$$\blacksquare y_{i+1} = y_{i+1}^* / \cosh\theta_i = y_i + x_i \tanh\theta_i$$

$$\blacksquare \text{Let } \theta_i = \tanh^{-1} 2^{-i} \text{ and } i = 1, 2, 3, \dots, N. (\because \tanh^{-1} 2^{-0} = \infty)$$

□ Hyperbolic CORDIC

$$x_{i+1} = x_i + d_i y_i 2^{-i}$$

$$y_{i+1} = y_i + d_i x_i 2^{-i}$$

$$z_{i+1} = z_i - d_i \theta_i$$

$$d_i = \operatorname{sign}(z_i)$$

$$\theta_i = \tanh^{-1} 2^{-i} \quad i = 1, 2, 3, \dots$$

Hyperbolic CORDIC – Rotation Mode(cont'd)

□ After n iterations, the recurrences of Hyperbolic CORDIC are as follows

■ $x_{n+1} \approx K'(x_1 \cosh z_1 + y_1 \sinh z_1)$

■ $y_{n+1} \approx K'(y_1 \cosh z_1 + x_1 \sinh z_1)$

□ Based on above equations, we can find e^x .

■ $e^x = \sinh x - \cosh x$

■ $z_1 = x, y_1 = 0, \text{ and } x_1 = 1$

□ Decomposition theorem:

■ Suppose σ is a real number and $\sigma_0 \geq \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$ is a finite sequence of real numbers such that

$$\sigma_i \leq \sigma_n + \sum_{j=i+1}^n \sigma_j, 0 \leq i \leq n \quad \text{and} \quad |\sigma| = \sum_{j=0}^n \sigma_j$$

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29

Hyperbolic CORDIC – Rotation Mode(cont'd)

□ Convergence problem

■ For example: Considering the case $i = n - 1$, it is contrary to the decomposition theorem.

$$\sigma_{n-1} > \sigma_n + \sigma_n = 2\sigma_n$$

■ On the other hand, the decomposition theorem is satisfied if iteration n were repeated.

$$\sigma_{n-1} \leq \sigma_n + 2\sigma_n = 3\sigma_n$$

■ Another evaluation

$$\tanh^{-1} 2^{-(i+1)} \geq \frac{1}{2} \tanh^{-1} 2^{-i}$$

Hyperbolic CORDIC – Rotation Mode(cont'd)

□ $x = 1, \cosh x = 1.543081 \dots, \sinh x = 1.175201 \dots$

□ Repeated at the
the following
iterations

- 4, 13, 40, ...,
 $i, 3i + 1, \dots$
- $g(k) = 3^{k+1}/2$
 $k = 1, 2, 3, \dots$

i	X _i	Y _i	Z _i	d _i
1	1.0011010100011110	0.0000000000000000	1.0000000000000000	1
2	1.0011010100011110	0.1001101010001111	0.0111001101100000	1
3	1.0101101111000010	0.1110011111010110	0.0011000111111101	1
4	1.0111100010111101	1.0001001101001111	0.0001000111010010	1
4	1.1000100111110010	1.0010101011011011	0.0000000111001101	1
5	1.1001110010011111	1.0100001101111010	-0.0000111000110111	-1
6	1.1001001010000100	1.0011011010010101	-0.0000011000110111	-1
7	1.1000110110101001	1.0011000001001011	-0.0000001000110111	-1
8	1.1000101101001001	1.0010110100101111	-0.0000000000110111	-1
9	1.1000101000011011	1.0010101110100100	0.0000000011001000	1
10	1.1000101010110001	1.0010110001101001	0.0000000001001000	1
11	1.1000101011111100	1.0010110011001100	0.0000000000001000	1
12	1.1000101100100010	1.0010110011111101	-0.0000000000010111	-1
13	1.1000101100001111	1.0010110011100100	-0.0000000000000111	-1
13	1.1000101100000110	1.0010110011011000	0.0000000000000000	-1
14	1.1000101011111100	1.0010110011001100	0.0000000000001000	1
15	1.1000101100000001	1.0010110011010010	0.0000000000000100	1
16	1.1000101100000011	1.0010110011010101	0.0000000000000010	1
	$\cosh x \approx 1.543$	$\sinh x \approx 1.175$		

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31

Hyperbolic CORDIC – Rotation Mode (cont'd)

□ The scaling constant K' depends on the repeated iterations,
and further can be derived as

$$\begin{aligned}
 K' &= \prod_{i=1}^n \frac{1}{\cosh \theta_i} \prod_{k=1}^m \frac{1}{\cosh \theta_{g(k)}} \\
 &= \prod_{i=1}^n \sqrt{1 - 2^{-2i}} \prod_{k=1}^m \sqrt{1 - 2^{-2g(k)}} \\
 &\approx 0.828\ 159\ 360 \dots
 \end{aligned}$$

- $g(1), g(1), \dots, g(m)$
- Special function in Hyperbolic CORDIC rotation mode
 - $e^z = \sinh z + \cosh z$
 - $\tanh z = \sinh z / \cosh z$
 - $w^t = e^{t \ln w}$

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32

Hyperbolic CORDIC – Vectoring Mode

- ❑ It is similar to use $\tan^{-1}x$ for $\sin x$ and $\cos x$ in trigonometric functions.
- ❑ The hyperbolic CORDIC using $\tanh^{-1}x$ to find $\sinh x$ and $\cosh x$

$$x_{i+1} = x_i + d_i y_i 2^{-i}$$

$$y_{i+1} = y_i + d_i x_i 2^{-i}$$

$$z_{i+1} = z_i - d_i \theta_i$$

$$d_i = -\text{sign}(y_i)$$

$$\theta_i = \tanh^{-1} 2^{-i} \quad i = 1, 2, 3, \dots$$

Hyperbolic CORDIC – Vectoring Mode (cont'd)

- ❑ Considering $x_1 = x$, $y_1 = y$, and $z_1 = 0$, we have the following CORDIC outputs after n iterations.

- $x_{n+1} \approx K' \sqrt{x^2 - y^2}$

- $y_{n+1} \approx 0$

- $z_{n+1} \approx \tanh^{-1} \left(\frac{y}{x} \right)$

- Range of $\tanh^{-1} \left(\frac{y}{x} \right)$: $[-1.12, 1.12]$

Hyperbolic CORDIC – Vectoring Mode (cont'd)

□ For example, $x_1 = 1$, $y_1 = y$, and $z_1 = 0$

i	X_i	Y_i	Z_i	d_i
1	0.0000011110101110	0.0000010100011110	0.0000000000000000	-1
2	0.0000010100011110	0.0000000101000111	0.1000110010011111	-1
3	0.0000010011001100	0.0000000000000000	0.1100111000000010	-1
4	0.0000010011001100	-0.0000000010011001	0.1110111000101101	1
4	0.0000010011000011	-0.0000000001001100	0.1101111000100111	1
5	0.0000010010111110	-0.0000000000000000	0.1100111000100010	1
6	0.0000010010111110	0.0000000000100101	0.1100011000100001	-1
7	0.0000010010111101	0.0000000000010010	0.1100101000100001	-1
8	0.0000010010111101	0.0000000000000100	0.1100110000100001	-1
9	0.0000010010111101	0.0000000000000010	0.1100110100100001	-1
10	0.0000010010111101	0.0000000000000001	0.1100110110100001	-1
11	0.0000010010111101	0.0000000000000000	0.1100110111100001	-1
12	0.0000010010111101	0.0000000000000000	0.1100111000000001	-1
13	0.0000010010111101	-0.0000000000000000	0.1100111000010001	1
13	0.0000010010111101	-0.0000000000000000	0.1100111000001001	1
14	0.0000010010111101	0.0000000000000000	0.1100111000000001	-1
15	0.0000010010111101	-0.0000000000000000	0.1100111000000101	1
16	0.0000010010111101	-0.0000000000000000	0.1100111000000011	1

$\tanh^{-1}(y/x) \approx 0.8047$

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35

Hyperbolic CORDIC

□ Special function in Hyperbolic CORDIC rotation mode

- $\ln w = 2 \tanh^{-1} \left[\frac{w-1}{w+1} \right]$
- $\sqrt{w} = \sqrt{(w + 1/4)^2 - (w - 1/4)^2}$
- $\cosh^{-1} w = \ln(w + \sqrt{1 - w^2})$
- $\sinh^{-1} w = \ln(w + \sqrt{1 + w^2})$

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36